

Advanced NLP

Monsoon '26

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Logistics

- Revised course
- Reading material
 - Papers, blogs, videos
- Strict deadlines. No exceptions for deadline extension



Topics

- Transformers (9 lectures)
 - Tokenization, Positional Embedding, Attention, Normalization, Layers, Activations, Optimizers, MoEs, Scaling
- Evaluation (3 lectures)
 - Statistics for Evaluation, Metrics, Benchmarking
- Decoding & Generation (4 lectures)
 - Traditional Strategies, Speculative Decoding, Parallel Decoding, Current Advances
- Quantization (1 lecture)



Topics

- Alternate Architectures (4 lectures)
 - RNN based architectures
 - State Space Models
 - Diffusion Language Models
- Reasoning in LLMs (4 lectures)
 - Reinforcement learning for NLP
 - DPO, PPO, GRPO
 - Reasoning in LLMs
- Whats Next (1 lecture)



Distribution of Grades

Item	Weightage
Project	50
4 Assignments	30 (6 + 7 + 7 + 10)
Mid Exam	10
Final Exam	10
Seminar *	10

Assignments (30%)

#	Release	Due	Grade
1	10 August	2 September	10 September
2	6 September	4 October	14 October
3	8 October	13 November	19 November
4	TBA	TBA	TBA



Exams (20%)

Mid Sem	As per institute calendar
End Sem	As per institute calendar

Seminar (10%)

- Replace one exam with a presentation in class
- Not more than **16** approved students
- **NOT** present one research paper but thorough presentation on a topic
- Evaluated on topic understanding, coverage, presentation and ability to answer questions
- Details on topics + form opens - post add/drop



Projects

Project (50%)

- Propose Custom Projects
- Team size: 3 or 4
- Random team allotment option available
- Resubmit proposal if rejected
- Utilize TA Office hours to discuss projects feasibility* and scope

* Compute cannot be provided :(



Project Expectation

- Advanced Level projects suitable to be submitted as papers
 - Highly encouraged to submit work to A* conferences/workshops
- Projects not suitable:
 - Finetuning/Instruction finetuning
 - RAG



Project Expectation

- Suitable Projects
 - Architecture optimization & changes
 - Novel Evaluation Methods
 - Datasets*



Project Areas

- Efficient/Low-Resource Methods
- Generation/Language Modeling
- Interpretability/Explainability
- Multilingual/Cross-lingual
- Multimodal
- NLP Applications
- Evaluation/Benchmarking
- Retrieval & Extraction
- Document Understanding
- Graph + LLMs
- Reasoning in LLM
- Diffusion Language Models



Project Timeline

Item	Deadline
Team Details/Assignment Form	6 August*
Interim Project Proposal	14 August
Proposal Rejection by TAs	16 August
Project Finalization, Mentor Assignment	20 August
Final Proposal	28 August
Mid Submission	30 September
Final Submission	31 October

* Institute Add/Drop August 5

Project Grading

Item	Weightage
Project Proposal	10
Mid Eval	20
Final Eval	20

Course Website



<https://ltrcnlp.github.io/M26/>